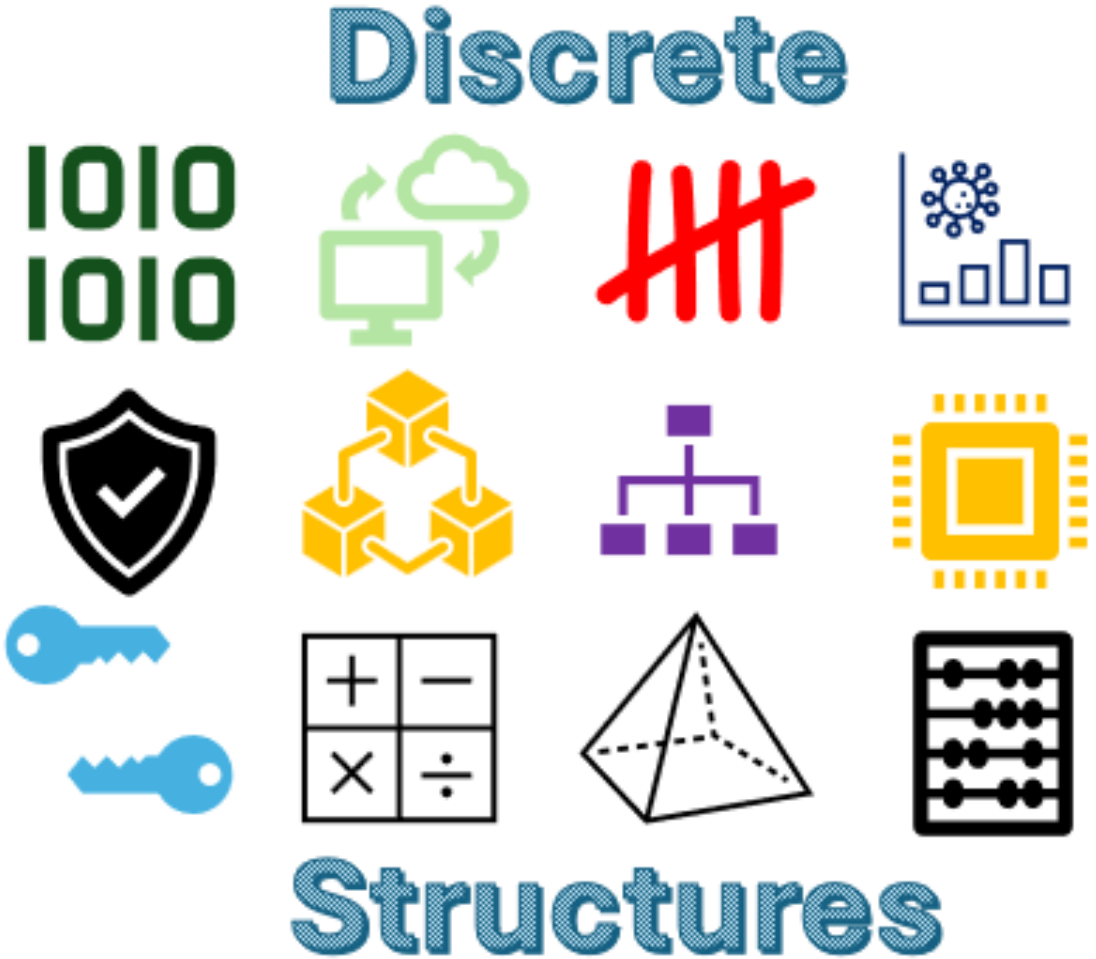
COMP 2310: Discrete Structures
Fall 2026

Chapter 1: Fundamentals



The addition principle (Inclusion-exclusion principle)

- Generalizing inclusion-exclusion principle for more than 2 variables

The addition principle: example

Example 10 A survey has been taken on methods of commuter travel. Each respondent was asked to check BUS, TRAIN, or AUTOMOBILE as a major method of traveling to work. More than one answer was permitted. The results reported were as follows: BUS, 30 people; TRAIN, 35 people; AUTOMOBILE, 100 people; BUS and TRAIN, 15 people; BUS and AUTOMOBILE, 15 people; TRAIN and AUTOMOBILE, 20 people; and all three methods, 5 people. How many people completed a survey form?

The addition principle: example

26. A survey of 500 television watchers produced the following information: 285 watch football games, 195 watch hockey games, 115 watch basketball games, 45 watch football and basketball games, 70 watch football and hockey games, 50 watch hockey and basketball games, and 50 do not watch any of the three kinds of games.

(a) How many people in the survey watch all three kinds of games?

(b) How many people watch exactly one of the sports?

1.3. Sequences

- A sequence is a list of objects arranged in a definite order
  - 1, 2, 3, 4
- Finite sequence
  - $n = 1, 2, 3, 4$
  - $1 \leq n \leq 4$
- Infinite sequence
  - $n = 1, 2, 3, 4, \ldots$
  - $1 \leq n$

1.3. Sequences: compact representation

- Recursive formula
  - Defines each term based on the previous one

Example 2 The list 3, 8, 13, 18, 23, . . . is an infinite sequence. The three dots in the expression mean "and so on," that is, continue the pattern established by the first few elements. ◆

$$a_1 = 3, a_n = a_{n-1} + 5, \ 2 \leq n.$$

- Explicit formula
  - Defines each term by using the position number

Example 3 Another infinite sequence is 1, 4, 9, 16, 25, . . . , the list of the squares of all positive integers. ◆

$$b_n = n^2, \ 1 \leq n.$$

1.3. Sequences: Characteristics function

- A = {110, 115, 121, 125, 126, 108, 105}

$$f_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A. \end{cases}$$

- Example 13

Example 13 Let $U = \{a, b, e, g, h, r, s, w\}$. The array of length 8 shown in Figure 1.17 represents U, since $A[k] = 1$ for $1 \leq k \leq 8$.

1	1	1	1	1	1	1	1

Figure 1.17

If $S = \{a, e, r, w\}$, then

$$f_S(x) = \begin{cases} 1 & \text{if } x = a, e, r, w \\ 0 & \text{if } x = b, g, h, s. \end{cases}$$

Hence the array in Figure 1.18 represents the subset S.

1	0	1	0	0	1	0	1

1.3. Sequences: Strings and regular expressions

- A = {a, b, c, d, e, ……, z}
- A* = consists of all finite sequences (words, strings) formed from A including the empty set

- Example 15

Example 15 Let $A = \{\text{John, Sam, Jane, swims, runs, well, quickly, slowly}\}$. Then A^* contains real sentences such as "Jane swims quickly" and "Sam runs well," as well as nonsense sentences such as "Well swims Jane slowly John." Here we separate the elements in each sequence with spaces. This is often done when the elements of A are words. ◆

1.3. Sequences: Strings and regular expressions

- A = {a, b, c, d, e, ……, z}
- M = {(,), ∗, ∨ }
- Regular expression: strings constructed from A and M
  - (): parenthesis
    - ab(bc)*: {ab, abbcbc, abbcbcbc, ...}
  - *: asterisk: zero or more repetition of the preceding element.
    - a*: {ε, a, aa, aaa, ...}
  - ∨: logical OR
    - a(b ∨ c): {ab, ac}

1.3. Sequences: Regular expression examples

$0^*(0 \vee 1)^*$	All sequences consisting of 0 or 1
$00^*(0 \vee 1)^*1$	Sequences starting with 0 and ending with 1
$(01)^*(01 \vee 1^*)$	String repeating 01 and ends with 1*

Reading

- Theorem 1

Environment setup

https://colab.research.google.com/

Colab notebook implementations

https://colab.research.google.com/drive/1nX6izoqyDHFIfIkcVbQcg4J7QoLVAI1J#scrollTo=Yo0FEYDvKDKQ